

Ideation Phase

Define the Problem Statements

Date	8 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	1 Marks

Scalability & Future Plan

The **Smart Agricultural Production** project is designed with a modular architecture, allowing it to be extended and enhanced in future versions. While the current system successfully recommends suitable crops based on soil and environmental parameters, it can be scaled to support a larger user base, additional agricultural services, and improved performance. This document outlines the current limitations, proposed scalability improvements, and the future development roadmap.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Application runs on a local Flask server only.	Limits accessibility for remote users.	High
2	No real-time weather or IoT sensor integration.	Recommendations depend on manually entered values.	High
3	The model uses a static training dataset.	Prediction accuracy may not reflect new agricultural trends.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for local or small-scale usage.	Deploy the application on a cloud platform such as Render, Railway, or AWS to support more users.
2	Data Storage	Uses a static dataset and trained model files.	Integrate a relational database (MySQL/PostgreSQL) for storing user records and prediction history.
3	Performance	Predictions are generated using a locally loaded model.	Optimize model loading, introduce caching, and improve backend processing for faster response times.

4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, secure API handling, and role-based access control in future versions.
---	----------	--	--

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
1	Complete Crop Recommendation Module	Day 1 – Day 3	Core functionality developed and tested.
2	Improve User Interface and Input Validation	Day 4 – Day 6	Better user experience and reliable input handling.
3	Integrate Machine Learning Model and Perform Testing	Day 7 – Day 9	Accurate crop prediction and stable application performance.
4	Documentation, Final Review, and Project Demonstration	Day 10	Complete project ready for evaluation and submission.